

Contribution Report

Team Enigma

Early Warning of Hydrate Formation in Offshore Well Service Lines

3W Dataset 2.0.0, Event 9

DLE-AI-202, Cohort I 2026 · Track 2 · AI Academy, Baku, Azerbaijan

<https://github.com/AMammedova/team-hydrate3w> (tag v1.0-final)

Member Contributions

Member	Code & Experiments	Report Sections	Slides
Mammadova, Aisel (M1)	<code>stats.py</code> , <code>availability.py</code> , <code>download.py</code> , annotated trace; cache sensitivity sweep	Sec. I (Introduction) & Sec. III (Dataset)	Slides 1–3
Huseynov, Shamistan (M2)	<code>splits.py</code> (two-population grouped CV), <code>fold_report()</code> , split validation tests	Sec. V (Experimental Setup)	Slide 4
Karimova, Rahima (M3)	<code>features.py</code> (95 features), <code>tune.py</code> , <code>calibrate.py</code> , reliability plot, feature ablation	Sec. IV-A (Baseline) & Sec. VI-A (XGBoost)	Slide 5
Mammadova, Gulnur (M4)	<code>tcn.py</code> , <code>gru.py</code> , <code>train_loop.py</code> , <code>profile.py</code> ; 12 A100 GPU runs	Sec. II (Related Work) & Sec. VI-B (Deep Models)	Slides 6–7
Arabova, Saida (M5)	<code>thresholds.py</code> , <code>metrics.py</code> , <code>aggregate.py</code> , <code>plots.py</code> , <code>run_all.sh</code>	Abstract, Sec. IV-C, VI-C, VII, VIII, & report editing	Slides 8–14

Contribution Descriptions

Mammadova, Aisel (M1) — Data Quality, Dataset Story & Early Slides. Aisel led the characterization of the 3W Event-9 dataset, uncovering that only 14 of the 57 real instances contain a transient phase. She implemented the sensor availability analysis (`availability.py`), dataset statistics (`stats.py`), and the annotated sensor trace figure. She authored Section I (Introduction) and Section III (Dataset) of the report, and prepared Slides 1–3 on problem motivation and dataset properties.

Huseynov, Shamistan (M2) — Cross-Validation Design & Experimental Setup. Shamistan designed and implemented the two-population grouped cross-validation strategy in `splits.py`. Because hydrate wells and normal-operation wells are completely disjoint, standard splitters fail to balance both populations; his solution pairs independent grouped splits across 3 folds. He authored Section V (Experimental Setup) of the report, implemented `fold_report()`, and prepared Slide 4 on the validation protocol.

Karimova, Rahima (M3) — Feature Engineering, XGBoost Baseline & Analysis. Rahima built the XGBoost pipeline: the 95-feature multi-timescale mask-aware extractor in `features.py`, grouped inner-CV tuning, Platt calibration, and feature ablation across 18 cells.

She authored Section IV-A (Feature-Engineered Baseline) and Section VI-A (XGBoost Performance) of the report, and prepared Slide 5 on feature engineering and baseline calibration.

Mammadova, Gulnur (M4) — Deep Architectures, GPU Training & Deep Results.

Gulnur implemented the Temporal Convolutional Network (TCN) and Gated Recurrent Unit (GRU) models along with the PyTorch training infrastructure (`fit()` with early stopping on validation PR-AUC, mixed precision, and seed determinism). She executed all 12 deep model training runs on the A100 GPU, authored Section II (Related Work) and Section VI-B (Deep Learning Results) of the report, and prepared Slides 6–7 on deep learning architectures.

Arabova, Saida (M5) — Evaluation Pipeline, Synthesis & Conclusion.

Saida developed the evaluation framework (`thresholds.py`, `metrics.py`, `aggregate.py`, `plots.py`, `run_all.sh`) implementing causal alarm logic at matched false-alarm budgets, event recall, and lead-time calculation. She authored the Abstract, Section IV-C (Causal Alarm Logic), Section VI-C (Head-to-Head Comparison), Section VII (Discussion & Limitations), and Section VIII (Conclusion & Tools), coordinated report editing and integration, and prepared Slides 8–14 covering evaluation results, operational trade-offs, and final takeaways.